

Pierre Mikhiel

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EDUCATION

New York University Abu Dhabi

Abu Dhabi, UAE

*B.A. in Interactive Media; Minors in Computer Science and Film & New Media – **Cum Laude** Graduated: May 2026*

Global Education: Completed 2 semesters abroad in **New York City** and **Shanghai**.

Relevant Coursework: VR Research and Applications, Designing Virtual Worlds, UX Design, Sound Technology, Data Structures, Algorithms, Virtual Body Performance.

RESEARCH EXPERIENCE

Machine Learning Research Assistant - ML5

Jan. – Current

Interactive Media NYUAD

Abu Dhabi, UAE

- Contribute to the development and creative application of ml5.js, an open-source, browser-based machine learning library built on TensorFlow.js.
- Design and prototype interactive, creative coding experiments that explore machine learning as a critical and artistic interface.
- Design a Machine Learning program that teaches ASL by recognizing hand gestures using HandPose.
- Develop visualizations and interfaces that communicate ML concepts to artists, educators, and students.

Research Assistant - VR Emotion Induction Toolkit

May – Sep. 2025

Aim Lab NYUAD

Abu Dhabi, UAE

- Contributed to a modular VR toolbox designed to systematically induce and study the six universal emotions and stress.
- Led the design of the Anger module, engineering interactive scenarios that use social unfairness and time pressure to evoke measurable emotional responses.
- Designed and implemented the 3D environments and animation sequences in Unity, crafting the visual pacing, movement, and interaction cues needed to reliably evoke targeted emotional states.

Co-Author | *vER: Virtual Human Companionship for Preoperative Care*

- Designed virtual agents that leverage social presence to mitigate patient anxiety in simulated pre-operative hospital settings.
- Validated the agents' efficacy by conducting a pilot study that integrated biometric tracking with qualitative post-study surveys.
- Presented research findings at prestigious international venues, including the IMX ACM Conference 2025 and EMRN Barcelona 2025.

PROJECTS & INSTALLATIONS

ML5 ASL Trainer - ML Research & Full-Stack Browser Application

- Proposed, led, and co-developed an interactive, browser-based ASL Trainer powered by computer vision in collaboration with Prof. Jack B. Du and Daniel Shiffman.
- Trained and verified the core gesture recognition dataset, embedded neural assets, and fine-tuned model performance for smooth, instant interactive feedback.
- Designed and deployed the end-to-end UI/UX across distinct A-Z/word modes, full-screen interactive layouts, progress metrics, and integrated reference visuals.

V!Real(ity) - Capstone Interactive Installation & Immersive VR Exhibition

- Designed and engineered a unified sensory ecosystem using an RGB DMX lighting setup, layered composite graphics, and context-dependent sound files to expose how reality is cognitively constructed.
- Developed branching interactive virtual reality experiences in Unity to study spatial priming, narrative heuristic bias, and player agency under environmental uncertainty.
- Configured real-time, multi-user shadow cast mechanics where collaborative physical positioning uncovered hidden visual layers, transforming isolated observation into shared social context.

Mondrian's Mirage - VR installation, Louvre Abu Dhabi

- Immersive VR environment inspired by Mondrian's compositions, allowing visitors to step inside shifting planes of color.

Whispers from the Depth - XR performance, NYUAD Black Box Theater

- XR dance performance where a motion-captured dancer's body is mapped to a virtual avatar projected behind her, appearing to dance on the souls of marine animals lost to pollution.

Beat-Hoven - Interactive music game (p5.js + Arduino), IM Showcase

- Playful music-making game that combines p5.js and Arduino, using beatboxing and physical interaction to build layered musical pieces in real time.

Alien Meta Rave - Live VR music performance, New York

- Live VR performance in which the audience sees only the performer's silhouette between projector and screen, dancing in a headset while a virtual rave world unfolds behind them.

Are You There? - Hybrid performance, La MaMa Studio (NYC)

- Hybrid live-digital performance blending projection, DJing, machine learning, and audience interaction to explore embodiment, virtual identity, and presence across physical and online space.

The Spiral - Large-scale sculpture, NYU New York

- Suspended wooden sculpture lit with RGB DMX fixtures arranged to cast overlapping colored shadows, transforming the form through shifting light and shadow.

Future of Rights of Nature - Sustainability diplomacy project, Swissnex

- Top 10 globally in a Swissnex-hosted sustainability diplomacy competition exploring environmental governance and Rights of Nature.

PRESENTATIONS & CONFERENCES

IMX ACM (2025) - "Virtual Companions in Preoperative VR"

EMRN Conference, Barcelona (2025) - "Affective Presence in Immersive Tech"

Louvre Abu Dhabi (2024) - VR immersive experience

ArtsIT2024 - Assisted in organizing the event, and covered the digital documentation including photography and media management

WORK EXPERIENCE

Internship

Nov. – Jan. 2025

Mental.ae

Abu Dhabi, UAE

- Performed QA testing on interactive food safety training modules, documenting key UX issues and informing design revisions that streamlined learner navigation.
- Used Cinemachine in Unity 3D to create dynamic camera systems and cinematic sequences that increased immersion in training simulations.

Residential Assistant

Aug. – Nov. 2023 & Aug. 2025 – Present

NYU Housing

Abu Dhabi

- Coordinated social and educational initiatives to support student wellness and build an inclusive residential community.

Fitness Assistant Coach

Jan. – May 2024

NYU Athletics Center

Shanghai

- Led group fitness activities and guided patrons on equipment safety to expand campus wellness engagement.

SKILLS & INTERESTS

Technical/Programming: C#, Python, JS, C++, Arduino, p5.js

Creative/Production Tools: Unity3D, Blender, DaVinci Resolve, Ableton, QLC+, OBS

Frameworks: XR Interaction Toolkit, FaceMesh, Arduino IDE

Other: Biometric integration (Motion capturing), Performance design, Creative coding

Languages: Arabic, English

Volunteering: Over 200 hours of volunteering in event management for international events

Interests: Passionate about designing emotionally resonant XR experiences by weaving together VR, kinetic hardware, Light, Sound, and interactive software to push the boundaries of immersive storytelling.